

Sales Analytics Pipeline using Azure Data Factory and Power BI

Professional Portfolio Report
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Project Objective

This project demonstrates an end-to-end cloud data analytics solution. It automates movement of multiple CSV files from Azure Blob Storage using Azure Data Factory and builds interactive Power BI dashboards for business reporting.

The main objective of this project is to build an automated data pipeline using **Azure Data Factory (ADF)**. The pipeline moves sales data from one location to another without any manual work. This helps businesses prepare their data for reporting and analysis.

In many companies, business data is stored in different files such as sales records, customer information, and product details. Employees often spend a lot of time downloading, uploading, and organizing these files before they can analyze the data. This manual process takes time and may lead to errors.

This project solves that problem by creating an automated cloud-based data pipeline.

The project uses **Azure Blob Storage** as the storage system, **Azure Data Factory** as the data integration tool, and **Power BI** as the reporting tool.

The objectives of this project are:

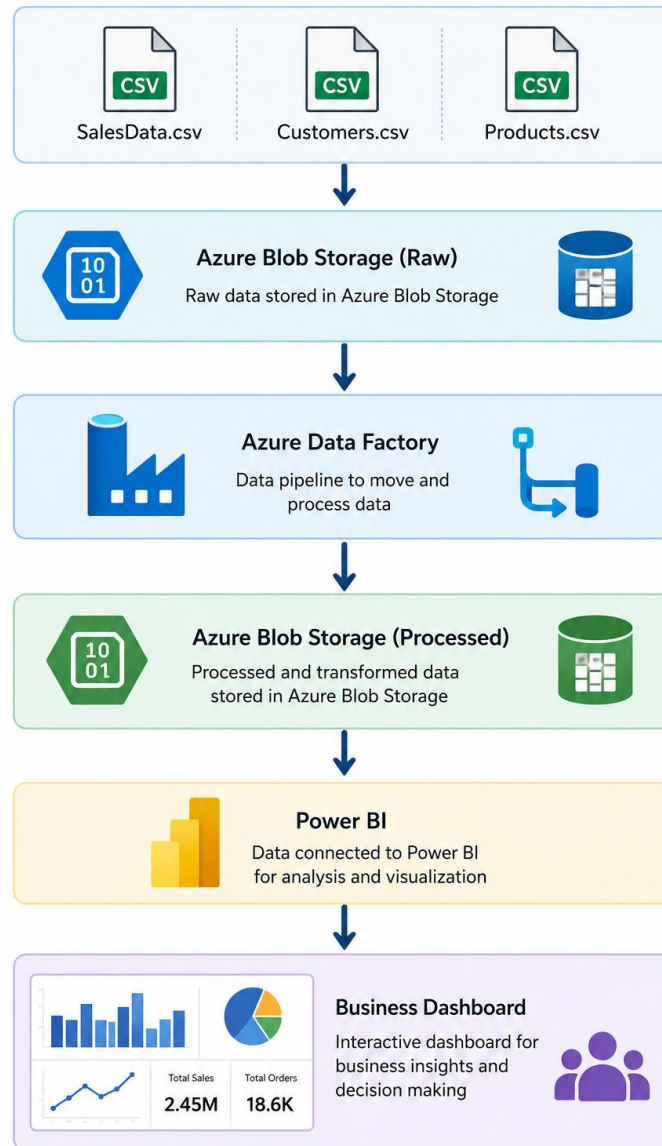
- Automate the movement of sales data using Azure Data Factory.
- Store raw and processed data securely in Azure Blob Storage.

- Eliminate manual file copying.
- Prepare clean and organized data for business reporting.
- Build interactive Power BI dashboards for business analysis.
- Learn cloud-based data engineering concepts.
- Demonstrate Azure Data Factory skills for a Data Analyst portfolio.
- Build a real-world project that is similar to enterprise data pipelines.

By completing this project, the entire process becomes faster, more reliable, and easier to manage.

Overall Architecture

The architecture shows how three CSV files move from the Raw container through Azure Data Factory into the Processed container before being analyzed in Power BI.



This project follows a simple cloud data pipeline architecture.

SalesData.csv

Customers.csv

Products.csv

|



Azure Blob Storage

(Raw Container)

|



Azure Data Factory

(Copy Activities)

|



Azure Blob Storage

(Processed Container)

|



Power BI



Business Dashboard

Architecture Explanation

1. Source Data (CSV Files)

The project starts with three CSV files.

- SalesData.csv
- Customers.csv
- Products.csv

These files contain business information such as customer details, products, sales amount, order dates, and product prices.

These files represent the raw data received by a company.

2. Azure Blob Storage (Raw Container)

The CSV files are uploaded into the Raw Data container.

The Raw container stores the original files exactly as they were received.

No changes are made to these files.

This allows businesses to keep a backup of the original data.

3. Azure Data Factory

Azure Data Factory is the heart of the project.

It connects to Azure Blob Storage using a Linked Service.

ADF reads the CSV files from the Raw container.

The pipeline contains three Copy Activities:

- Copy Sales Data
- Copy Customers Data
- Copy Products Data

Each Copy Activity moves one CSV file from the Raw container to the Processed container.

The pipeline can also be scheduled to run automatically every day using a Trigger.

4. Azure Blob Storage (Processed Container)

The processed container stores the files after they have been copied by Azure Data Factory.

These files become the source for reporting and analytics.

The processed data is organized and ready for business users.

5. Power BI Dashboard

Power BI connects to the processed CSV files.

The data model is created using relationships between Sales, Customers, and Products.

Interactive dashboards are built to help users understand business performance.

The dashboards provide insights such as:

- Total Sales
- Monthly Revenue
- Sales by Product
- Top Customers
- Sales by State

Managers can use these dashboards to make better business decisions.

Pipeline Flow

The pipeline is the workflow that moves data automatically.

The complete pipeline follows these steps.

Step 1

Business data is generated.

Three CSV files are available:

- SalesData.csv
 - Customers.csv
 - Products.csv
-

Step 2

All CSV files are uploaded into the **Raw Data** container inside Azure Blob Storage.

These files become the source of the pipeline.

Step 3

Azure Data Factory starts the pipeline.

The pipeline connects to Azure Blob Storage using the Linked Service.

Step 4

ADF reads the source datasets.

Three datasets are used.

- DS_Sales
- DS_Customers
- DS_Products

Step 5

Three Copy Activities are executed.

Activity 1

Copies SalesData.csv

Raw → Processed

Activity 2

Copies Customers.csv

Raw → Processed

Activity 3

Copies Products.csv

Raw → Processed

Step 6

The processed files are stored in the Processed Data container.

These files are clean and ready for reporting.

Step 7

Power BI connects to the processed files.

Relationships are created between the tables.

Measures and calculations are added.

Step 8

Interactive dashboards are created.

The dashboards display:

- Total Sales

- Total Orders
- Total Customers
- Sales by Product
- Monthly Revenue Trend
- Top Customers
- Sales by State

This completes the end-to-end cloud analytics pipeline.

Dataset Details

The project uses three datasets.

Dataset 1 – SalesData.csv

This is the main transaction table.

It stores all sales records.

Important Columns

- OrderID
- CustomerID
- ProductID
- OrderDate
- Quantity
- SalesAmount

Purpose:

This dataset is used to calculate:

- Total Sales
- Monthly Revenue
- Sales Trend

- Customer Revenue
 - Product Revenue
-

Dataset 2 – Customers.csv

This table stores customer information.

Important Columns

- CustomerID
- CustomerName
- City
- State
- CustomerType

Purpose:

This dataset helps identify:

- Top Customers
 - Customer Locations
 - Customer Categories
 - Customer Revenue
-

Dataset 3 – Products.csv

This table stores product information.

Important Columns

- ProductID
- ProductName
- Category
- Brand
- UnitPrice

Purpose:

This dataset is used to analyze:

- Product Sales
- Product Revenue
- Best Selling Products
- Product Categories

Relationships

The tables are connected using common keys.

SalesData.CustomerID → Customers.CustomerID

SalesData.ProductID → Products.ProductID

These relationships allow Power BI to combine information from all three tables and create meaningful reports.

Project Results

The project successfully created a complete cloud-based data pipeline using Azure services.

Key Achievements

- Successfully created Azure Storage Account.
- Created Raw and Processed Blob Storage containers.
- Uploaded all source CSV files.
- Created Linked Service in Azure Data Factory.
- Created datasets for all CSV files.
- Developed a pipeline with three Copy Activities.
- Successfully copied files from the Raw container to the Processed container.
- Configured a schedule trigger for automatic execution.
- Monitored successful pipeline execution using Azure Monitor.
- Connected processed data to Power BI.
- Created interactive dashboards for business reporting.
- Published the complete project to GitHub.

Business Results

The project provides several business benefits.

Before Automation	After Automation
Manual file upload	Automatic data movement
Slow reporting	Faster reporting
Multiple manual steps	Single automated pipeline
Higher chance of errors	Reliable process
Difficult monitoring	Azure Monitor tracks every run
Manual data preparation	Ready-to-use processed data

Final Outcome

This project demonstrates how Azure Data Factory can automate data movement between cloud storage containers and prepare data for business reporting.

Using Azure Blob Storage, Azure Data Factory, and Power BI together creates a complete end-to-end cloud analytics solution.

The project improves efficiency, reduces manual work, minimizes errors, and enables faster business decision-making through interactive dashboards.

It also demonstrates practical skills in cloud data engineering, data integration, automation, and business intelligence, making it a strong portfolio project for Data Analyst and Azure Data Factory roles.

Skills Learned

This project helped me learn how to build a complete cloud-based data pipeline using Microsoft Azure services. I gained practical experience in storing data, moving data automatically, monitoring pipeline execution, and creating interactive business dashboards.

Azure Blob Storage

- Created an Azure Storage Account to store business data securely in the cloud.
- Created separate containers for Raw Data and Processed Data.
- Uploaded and managed CSV files inside Blob Storage.
- Learned how cloud storage is used in real-world data engineering projects.

Azure Data Factory (ADF)

- Created Linked Services to connect Azure Data Factory with Azure Blob Storage.
- Built datasets for source and destination CSV files.
- Designed an automated pipeline using three Copy Activities.
- Configured source, sink, and column mapping.
- Validated and published the pipeline successfully.
- Learned how to automate repetitive data movement tasks.

Pipeline Monitoring and Scheduling

- Created a Schedule Trigger to run the pipeline automatically every day.

- Monitored pipeline execution using the Monitor page.
- Verified successful pipeline runs and activity status.
- Understood the importance of monitoring and troubleshooting cloud pipelines.

Power BI

- Connected Power BI to the processed data stored in Azure Blob Storage.
- Built relationships between Sales, Customers, and Products tables.
- Created measures to calculate Total Sales, Orders, Customers, and Revenue.
- Designed interactive dashboards for business reporting and analysis.
- Learned how data visualization helps businesses make better decisions.

GitHub

- Published the complete project to GitHub.
- Organized project files, screenshots, documentation, and datasets.
- Learned how version control helps manage and share projects professionally.

Overall Learning

By completing this project, I improved my understanding of cloud computing, data integration, automation, business intelligence, and reporting. I also gained practical experience with Azure services that are commonly used by data analysts and data engineers in real business environments.

Business Benefits

This automated cloud solution provides several important benefits to businesses.

Faster Data Processing

The pipeline automatically copies business data from the Raw container to the Processed container. This saves time and allows reports to be prepared much faster than manual methods.

Reduced Manual Work

Employees no longer need to download, upload, or copy files every day. The automated pipeline performs these tasks without human intervention, allowing employees to focus on more valuable work.

Improved Data Accuracy

Automation reduces the possibility of human errors such as missing files, duplicate data, or incorrect uploads. This improves the quality and reliability of business reports.

Better Decision Making

Power BI dashboards provide clear insights into sales performance, customer behavior, product revenue, and regional sales. Managers can use these insights to make faster and more informed business decisions.

Scalable Solution

As the business grows, additional files, datasets, or pipelines can easily be added without changing the overall architecture. This makes the solution suitable for both small and large organizations.

Secure Cloud Storage

Azure Blob Storage stores business data securely in the cloud with built-in security and access controls. This helps protect important business information.

Easy Monitoring

Azure Data Factory Monitor allows users to track every pipeline run, view execution history, and quickly identify any issues. This improves operational reliability and reduces downtime.

Reusable Architecture

The same pipeline design can be reused for other business processes such as inventory management, finance reporting, HR data processing, or customer analytics. This saves development time for future projects.

Cost and Time Savings

By automating repetitive tasks, businesses reduce operational costs, improve employee productivity, and generate reports more quickly. This leads to better efficiency and improved business performance.

Conclusion

The **Sales Analytics Pipeline using Azure Data Factory** successfully demonstrates how cloud technologies can automate business data processing and improve reporting. The project combines Azure Blob Storage, Azure Data Factory, and Power BI to create a complete end-to-end analytics solution.

The automated pipeline moves multiple CSV files from the Raw container to the Processed container without manual work. This reduces errors, saves time, and prepares clean data for business reporting. Power BI then transforms the processed data into interactive dashboards that help users understand sales performance, customer trends, product revenue, and regional sales.

This project also demonstrates practical skills in cloud storage, data integration, pipeline automation, monitoring, scheduling, and business intelligence. These are important skills for Data Analyst, Business Intelligence Analyst, and Azure Data Factory roles.

Overall, this project provides a simple, reliable, and scalable cloud solution for automating data movement and creating meaningful business insights. It serves as a strong portfolio project that demonstrates real-world experience with Microsoft Azure and modern data analytics tools.

Work Screenshots

Azure Storage Account

Azure Storage Account stores all business files securely in cloud containers.

The screenshot displays the Microsoft Azure portal interface. The top navigation bar includes the 'Microsoft Azure' logo, an 'Upgrade' button, a search bar, and user profile information for 'vishal30prajapati@ou'. The left sidebar contains a navigation menu with options like 'Home', 'Dashboard', 'All services', and 'FAVORITES'. The main content area shows the 'Overview' page for a storage account named 'salesanalyticsstorage1'. This page includes a search bar, action buttons (Upload, Open in Explorer, Delete, Move, Refresh, Open in mobile, CLI / PS, Feedback), and a table of account details. Below the details table, there are tabs for 'Properties', 'Monitoring', 'Capabilities (7)', 'Recommendations (4)', 'Tutorials', and 'Tools + SDKs'. The 'Properties' tab is active, showing settings for 'Blob service' and 'File service'. The 'Security' and 'Networking' sections are also visible on the right side of the page.

Essentials	
Resource group (move)	: Vishal
Location	: southeastasia
Subscription (move)	: Azure subscription 1
Subscription ID	: 5d066811-5d80-4e55-9e33-f3838a0ba55b
Disk state	: Available
Tags (edit)	: Add tags
Performance	: Standard
Replication	: Locally redundant storage (LRS)
Account kind	: StorageV2 (general purpose v2)
Provisioning state	: Succeeded
Created	: 22/06/2026, 11:27:23

Blob service	
Hierarchical namespace	Disabled
Default access tier	Hot
Blob anonymous access	Disabled
Blob soft delete	Enabled (7 days)
Container soft delete	Enabled (7 days)
Versioning	Disabled
Change feed	Disabled
NFS v3	Disabled
Allow cross-tenant replication	Disabled
Storage tasks assignments	None

File service	
Identity-based access	Not configured
Default share-level permissions	Disabled
Soft delete	Enabled (7 days)

Security	
Require secure transfer for REST API operations	Enabled
Storage account key access	Enabled
Minimum TLS version	Version 1.2
Infrastructure encryption	Disabled

Networking	
Public network access	Enabled
Public network access scope	Enabled from all networks
Private endpoint connections	0
Network routing	Microsoft network routing
Endpoint type	Standard

Blob Storage Containers

Separate containers improve data organization and simplify data pipelines.

Create a resource

Home

Dashboard

All services

FAVORITES

Application Templates

All resources

Resource groups

App Services

Function App

Azure SQL Database

Azure Cosmos DB

Virtual machines

Load balancers

Storage accounts

Virtual networks

Microsoft Entra ID

Monitor

Advisor

Microsoft Defender for Cloud

Cost Management + Billing

Help + support

Overview

Activity log

Tags

Diagnose and solve problems

Access Control (IAM)

Data migration

Events

Storage browser

Storage Mover

Partner solutions

Resource visualizer

Data storage

Containers

Classic file shares

Queues

Tables

Security + networking

Networking

Front Door and CDN

Access keys

Home > salesanalyticsstorage1

salesanalyticsstorage1

Storage account

Containers

Search

Add container

Upload

Refresh

Delete

Change access level

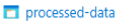
Restore containers

Edit columns

Search containers by prefix

Only show active containers

Showing all 3 items

☐	Name	Last modified	Anonymous access level	Lease state	Storage connector
☐	 logs	22/06/2026, 11:27:47	Private	Available	-
☐	 processed-data	22/06/2026, 11:28:52	Private	Available	-
☐	 raw-data	22/06/2026, 11:28:36	Private	Available	-

Raw Data Container

Original CSV files uploaded by users are stored here.

Create a resource

Home

Dashboard

All services

FAVORITES

Application Templates

All resources

Resource groups

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Function App

Azure SQL Database

Azure Cosmos DB

Virtual machines

Load balancers

Storage accounts

Virtual networks

Microsoft Entra ID

Monitor

Advisor

Microsoft Defender for Cloud

Cost Management + Billing

Help + support

Home > salesanalyticsstorage1 | Containers

raw-data

Container

Search

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Add Directory

Upload

Change access level

Refresh

Delete

Copy

Paste

Rename

Acquire lease

Break lease

Edit columns

raw-data

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 3 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state
☐	Customers.csv	22/06/2026, 11:29:39	Hot (Inferred)	Block blob	217.52 KiB	Available
☐	Products.csv	22/06/2026, 11:29:38	Hot (Inferred)	Block blob	182.76 KiB	Available
☐	SalesData.csv	22/06/2026, 11:29:38	Hot (Inferred)	Block blob	188.18 KiB	Available

Processed Data Container

ADF copies processed files into this container for reporting.

Microsoft Azure

Upgrade

Search resources, services, and docs (G+)

Copilot

vishal30prajapati@outlo

DEFAULT DIRECTORY (VISHAL30

Create a resource

Home

Dashboard

All services

FAVORITES

Application Templates

All resources

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Storage accounts

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Microsoft Entra ID

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Microsoft Defender for Cloud

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Help + support

Home > salesanalyticsstorage1 | Containers

processed-data

Container

Search

+ Add Directory

Upload

Change access level

Refresh

Delete

Copy

Paste

Rename

Acquire lease

Break lease

Edit columns

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

processed-data

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 3 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state
☐	Customers.csv	25/06/2026, 09:00:23	Hot (Inferred)	Block blob	217.52 KiB	Available
☐	Products.csv	25/06/2026, 09:00:22	Hot (Inferred)	Block blob	182.76 KiB	Available
☐	SalesData.csv	25/06/2026, 09:00:21	Hot (Inferred)	Block blob	188.18 KiB	Available

Linked Service Configuration

The Linked Service securely connects Azure Data Factory to Azure Blob Storage.

The screenshot displays the Microsoft Azure portal interface for configuring a linked service. The left sidebar shows the navigation menu with categories like General, Connections, Source control, Author, Security, and Workflow orchestration manager. The main area is titled 'Linked services' and shows a table with one item, 'LS_BlobStorage', of type 'Azure Blob Storage'. The right pane is titled 'Edit linked service' and contains the configuration details for 'LS_BlobStorage'.

Linked services

Linked service defines the connection information to a data store or compute. [Learn more](#)

+ New

Filter by name Annotations : Any

Showing 1 - 1 of 1 items

Name	Type	Rel
LS_BlobStorage	Azure Blob Storage	6

Edit linked service

Azure Blob Storage [Learn more](#)

To avoid publishing immediately to Data Factory, please use Azure Key Vault to retrieve secrets securely. [Learn more](#)

Name *
LS_BlobStorage

Description

Connect via integration runtime *
AutoResolveIntegrationRuntime

Authentication type
Account key

Connection string Azure Key Vault

Account selection method
From Azure subscription Enter manually

Storage account name *
salesanalyticstorage1

Storage account key *

Partitioned DNS enabled ☐

Endpoint suffix
core.windows.net

Additional connection properties

Apply Cancel

Connection successful
Test connection

Datasets

Datasets define the source and destination files used by the pipeline.

The screenshot displays the Microsoft Fabric Data Factory interface. On the left, the 'Factory Resources' sidebar shows a tree view with categories: Pipelines (1), Change Data Capture (preview) (0), Datasets (6), Data flows (0), Power Query (0), and Templates (0). Under the 'Datasets' category, several datasets are listed, including 'DS_Sales_Processed', which is currently selected. The main workspace shows the configuration for 'DS_Sales_Processed' under the 'Connection' tab. The configuration includes a linked service 'LS_BlobStorage', a file path 'processed-data / Directory / File name', and various options for compression, delimiters, encoding, quote and escape characters, and a checked 'First row as header' option. A 'Null value' field is also present. The top of the interface shows a navigation bar with 'main branch', 'Validate all', 'Save all', 'Publish', and a 'Migrate to Fabric (Preview)' button.

Factory Resources

- Pipelines 1
- Change Data Capture (preview) 0
- Datasets 6
 - DS_Customers
 - DS_Customers_Processed
 - DS_Products
 - DS_Products_Processed
 - DS_Sales
 - DS_Sales_Processed**
- Data flows 0
- Power Query 0
- Templates 0

Connection Schema Parameters

Linked service * LS_BlobStorage [Test connection](#) [Edit](#) [New](#) [Learn more](#)

File path * processed-data / Directory / File name [Browse](#) [Preview data](#) [Detect format](#)

Compression type No compression

Column delimiter Comma (,)

Row delimiter Default (\r,\n, or \r\n)

Encoding Default(UTF-8)

Quote character Double quote (")

Escape character Backslash (\)

First row as header ☒

Null value

ADF Pipeline

Three Copy Activities move Customers, Products and Sales data automatically.

main branch

Validate all

Save all

Publish

Migrate to Fabric (Preview)

Factory Resources

Filter resources by name

Pipelines1

SalesAnalyticsPipeline

Change Data Capture (preview)0

Datasets6

Data flows0

Power Query0

Templates0

SalesAnalyticsPipe...

Activities

Search activities

Move and transform

Synapse

Azure Data Explorer

Azure Function

Batch Service

Databricks

General

HDInsight

Iteration & conditionals

Machine Learning

Power Query

Saved

Save as template

Validate

Validate copy runtime

Debug

Trigger (1)

Copy data

Copy data1

Copy data

Copy data2

Copy data

Copy data3

General

Source

Sink

Mapping

Settings

User properties

Name *

Copy data1

Description

Activity state

Activated

Deactivated

Timeout

0.12:00:00

Retry

0

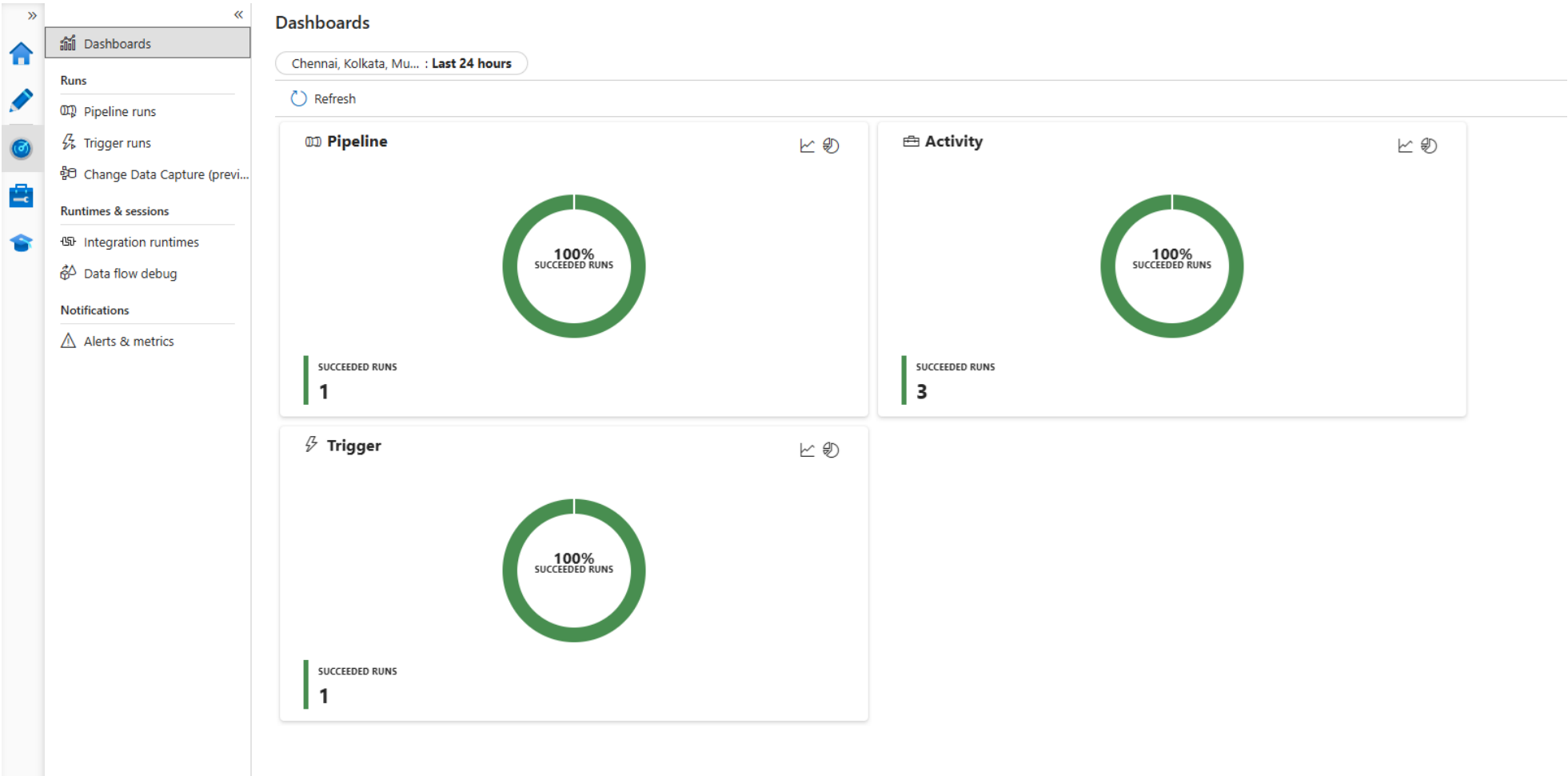
Retry interval (sec)

30

Secure output

Pipeline Monitoring

The Monitor page confirms successful execution and activity status.



Pipeline Trigger

A daily trigger automates pipeline execution every morning.

>>

<<

Home

Dashboard

Pipeline runs

Trigger runs

Change Data Capture (previ...

Runtimes & sessions

Integration runtimes

Data flow debug

Notifications

Alerts & metrics

Trigger runs

All

Schedule

Tumbling window

Storage events

Custom events

Refresh

Edit columns

Chennai, Kolkata, Mu... : Last 24 hours

Trigger name : Daily Trigger

Status : All

Runs : Latest runs

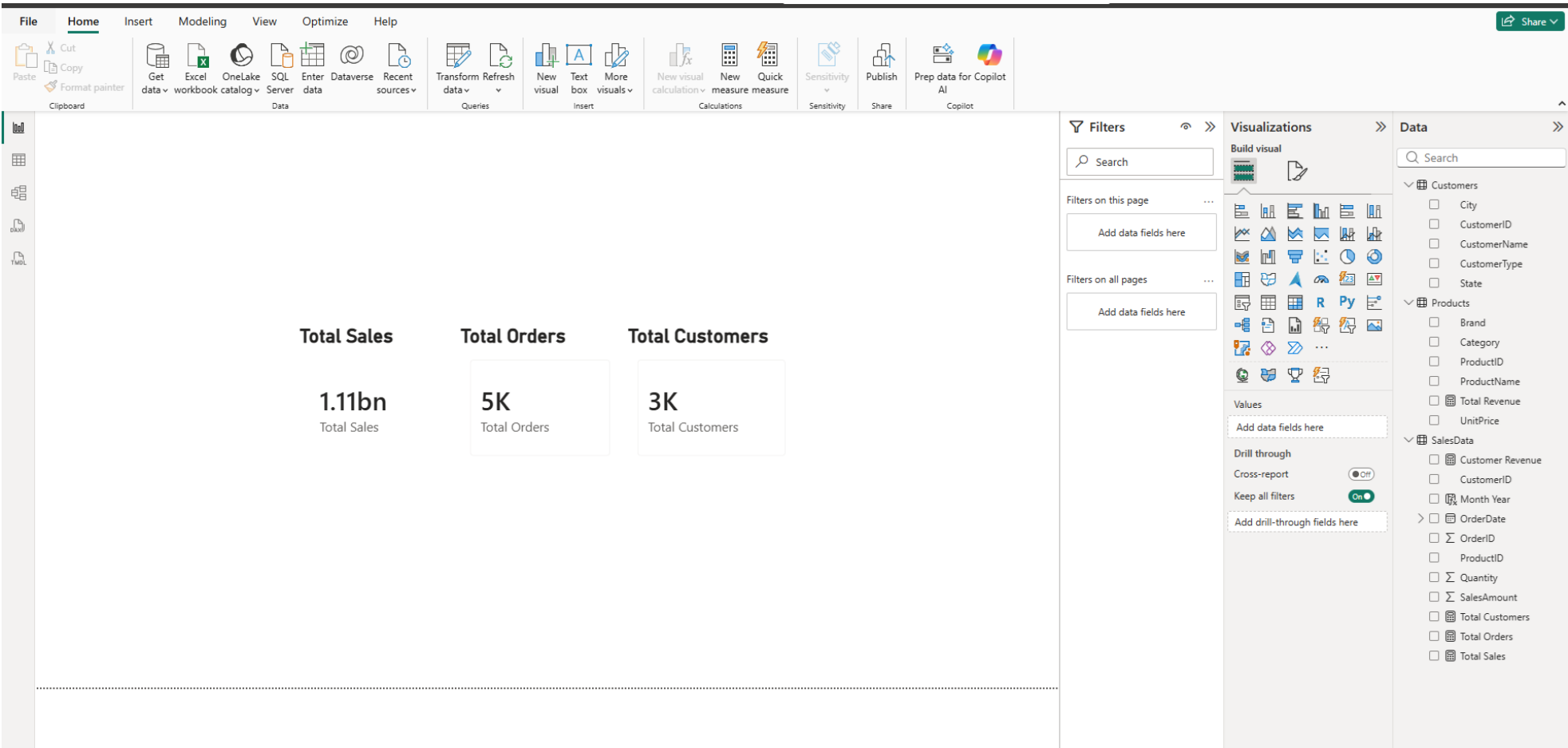
Export to CSV

Showing 1 - 1 items

Trigger name ↑↓	Trigger type	Trigger time ↑↓	Status ↑↓	Pipelines	Run	Message	Properties	Run ID
Daily Trigger	Schedule trigger	6/25/2026, 9:00:00 AM	Succeeded	1	Original			085841924868539788695...

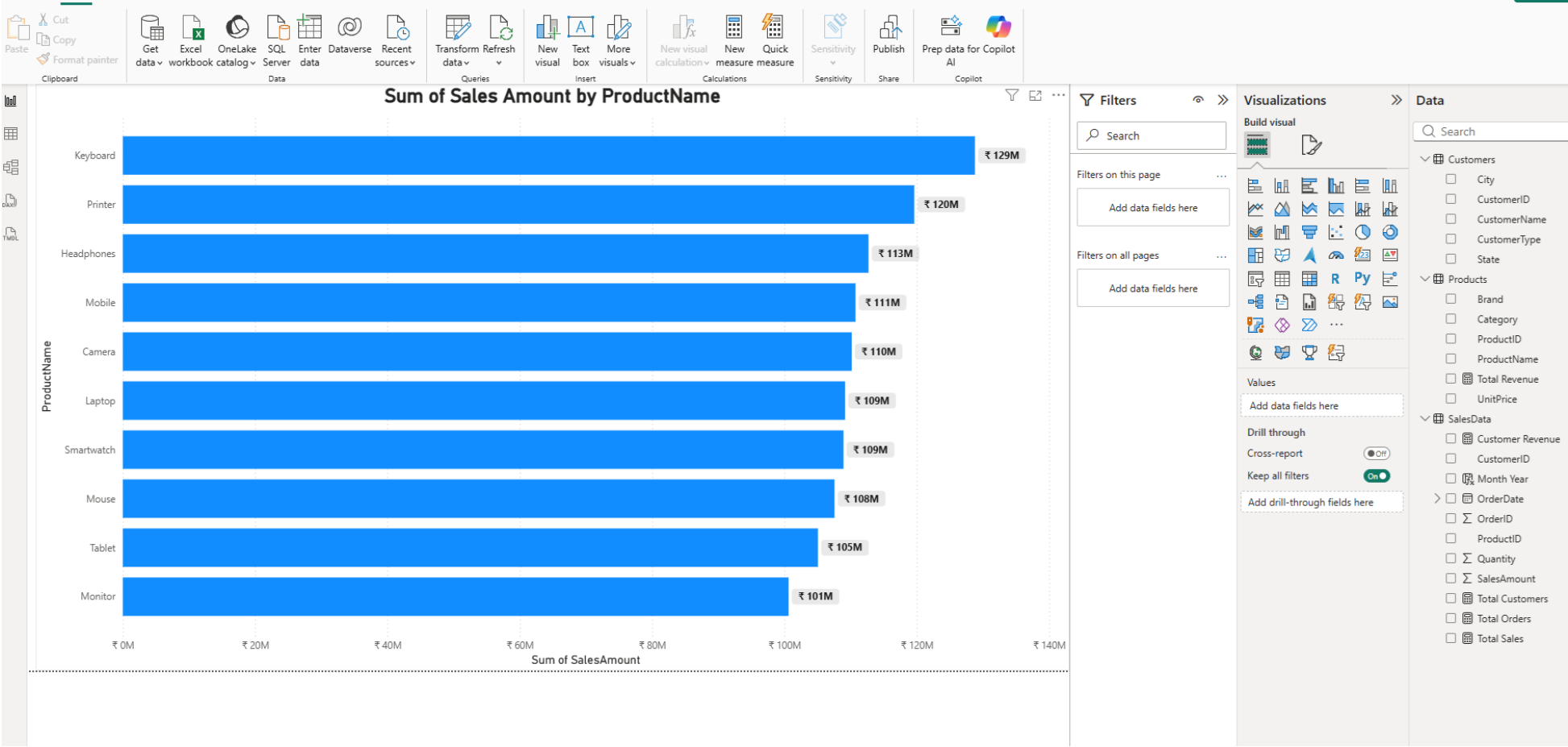
Dashboard 1 - Sales Overview

Executive KPI dashboard displaying Total Sales, Orders and Customers.



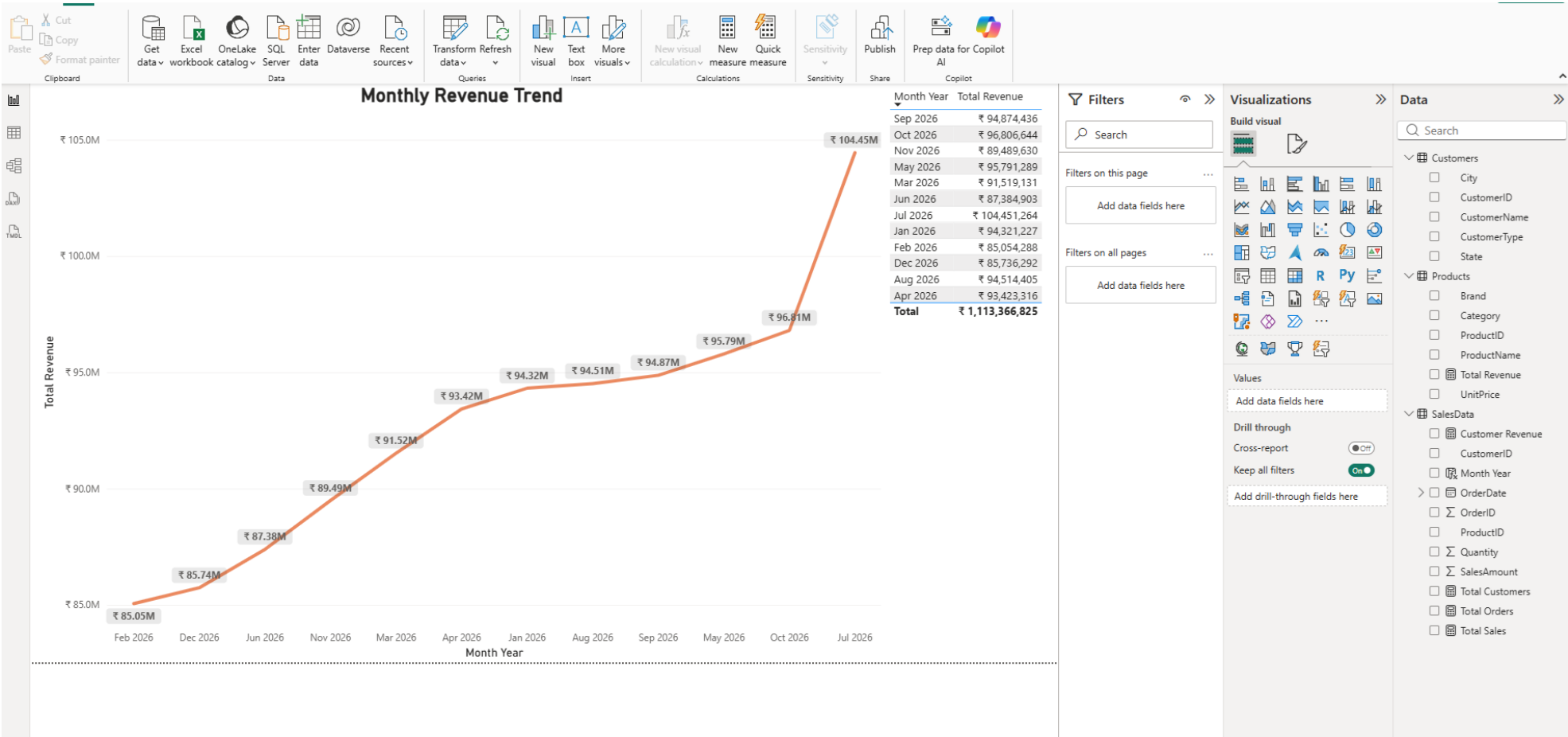
Dashboard 2 - Sales by Product

Compares revenue generated by each product category.



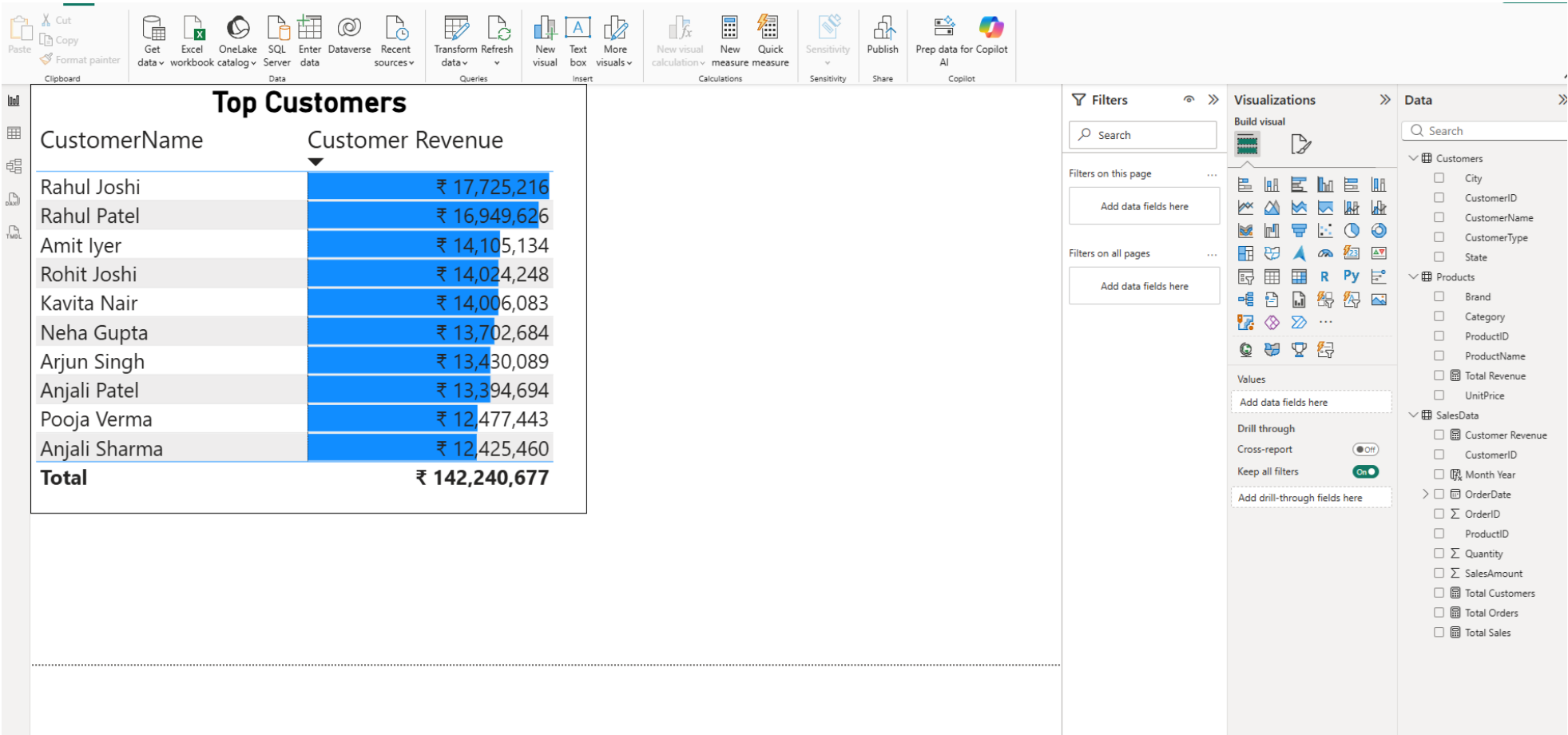
Dashboard 3 - Monthly Revenue

Shows monthly revenue trend for business performance analysis.



Dashboard 4 - Top Customers

Ranks customers based on generated revenue.



Dashboard 5 - Sales by State

Displays geographical sales distribution using a map.

